

Artificial Intelligence — Lecture 6

Advanced AI Sub-Domains: Machine Learning, NLP, and Computer Vision

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Lecture 6 at a Glance

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Advanced AI Sub-Domains

Course Learning Outcome

CLO6: Categorize advanced AI sub-domains such as Machine Learning, Natural Language Processing, and Computer Vision.

Lecture purpose

- Explain how **Machine Learning** allows agents to improve from data.
- Distinguish supervised, unsupervised, and reinforcement learning settings.
- Describe how **Natural Language Processing** represents and processes text.
- Describe how **Computer Vision** converts pixels into semantic information.
- Compare these domains by input type, output type, learning objective, and common applications.

Conceptual structure

Stage	Core question
Machine Learning	How can a system improve from data rather than explicit rules?
NLP	How can a system represent, interpret, and generate language?
Computer Vision	How can a system extract meaning from images and video?
Integration	How do modern AI systems combine these sub-domains?

💡 Main idea: advanced AI sub-domains often use learning-based models, but they differ in the kind of data they process and the tasks they solve.

Learning Objectives

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By the end of Lecture 6, you will be able to:

1. Define **Machine Learning** and explain why learning is central to modern AI.
2. Distinguish **supervised, unsupervised, and reinforcement learning**.
3. Formalize a supervised learning problem using examples, hypotheses, **loss, and generalization**.
4. Explain **overfitting, underfitting**, and the need for train/validation/test evaluation.
5. Describe common **NLP representations** such as tokens, vocabularies, embeddings, and language models.
6. Explain the intuition behind **attention** and transformer-based language processing.
7. Describe how computer vision systems process images using **features and convolutional neural networks**.
8. Categorize AI applications according to their **data type, learning setting, and output type**.

Lecture Roadmap

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From learning from data to language and perception

Stage	Main idea	Student output
1. Machine Learning	Models learn regularities from examples and use them to make predictions or decisions.	Identify the learning setting and formalize inputs, targets, model, and loss.
2. Natural Language Processing	Text must be transformed into numerical representations before models can reason over it.	Explain tokenization, embeddings, language modeling, and attention at a conceptual level.
3. Computer Vision	Images are numerical arrays whose spatial structure must be exploited.	Explain the role of convolution, feature maps, and task-specific visual outputs.
4. Synthesis	AI domains differ but share data, representations, models, evaluation, and limitations.	Categorize real AI systems by domain and learning approach.

Part I

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Machine Learning

Machine Learning shifts AI from manually specified behavior to models that **improve through experience**.

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What Is Machine Learning?

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Learning means improving performance from experience

A system is said to learn when its **performance on a task improves** as it receives more experience or data.

A classic learning formulation links three components:

experience E , task T , performance measure P

A spam filter, for example, improves at classifying email messages after observing many **labeled examples** of spam and non-spam messages.

💡 Machine Learning is **not a replacement for the rest of AI**. It is a way to construct models that can be used inside agents, search systems, language systems, vision systems, and decision-making systems.

Why Learning Became Central in AI

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Some knowledge is too complex to write as rules

Rule-based difficulty

For many tasks, it is difficult to **manually specify all rules**:

- recognizing faces under different lighting conditions,
- understanding informal natural language,
- detecting fraudulent behavior,
- recommending items to users,
- controlling robots in changing environments.

Learning-based idea

Instead of writing every rule, we provide **examples** and let the model estimate **useful patterns**.

A vision model does not receive a hand-written definition of “cat.” It learns **visual regularities** from labeled images.

Learning from data also introduces risks: **bias**, **overfitting**, poor data quality, and limited reliability outside the training distribution.

Learning Problem Formalization

Supervised learning as a mathematical object

A supervised learning dataset is often written as

$$D = \{(x_i, y_i)\}_{i=1}^m$$

where x_i is an **input example** and y_i is the **desired output**.

A model or hypothesis **maps inputs to predicted outputs**:

$$h_\theta : X \rightarrow Y$$

The learning objective is to choose parameters θ that **minimize prediction error**, often expressed as empirical risk:

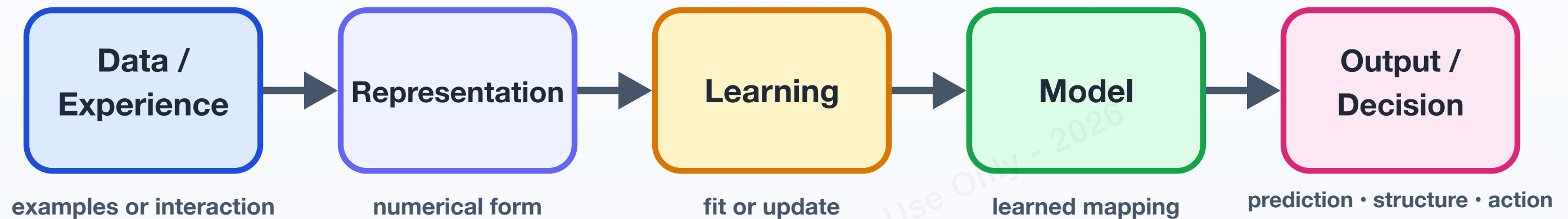
$$\theta^* = \arg \min_{\theta} \frac{1}{m} \sum_{i=1}^m L(h_\theta(x_i), y_i)$$

The **loss function** L measures how bad a prediction is. Learning is the process of **reducing this loss** while preserving the ability to **generalize**.

Machine Learning Pipeline

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From data or experience to improved performance



Learning uses data or experience to improve performance on a task.

Different learning settings use different sources of experience and different objectives.

💡 **Key idea.** Supervised learning learns from labeled examples, unsupervised learning discovers structure in data, and reinforcement learning improves behavior through interaction and reward.

Main Types of Machine Learning

The learning setting depends on the available feedback

Learning type	Data form	Goal	Example
Supervised learning	Inputs with known labels or target values	Learn a mapping from X to Y	Image classification, spam detection, price prediction
Unsupervised learning	Inputs without explicit labels	Discover structure or representations	Clustering customers, dimensionality reduction
Reinforcement learning	Interaction with an environment and rewards	Learn a policy for sequential decisions	Game playing, robot control, adaptive recommendation

💡 The key diagnostic question is: **what kind of feedback does the learner receive?**

Supervised Learning

Classification and regression

Classification

The output belongs to a **finite set of categories**:

$$Y = \{c_1, c_2, \dots, c_k\}$$

Examples:

- spam or not spam,
- disease category,
- object class in an image,
- sentiment: positive, neutral, negative.

Regression

The output is **numerical**:

$$Y \subseteq \mathbb{R}$$

Examples:

- house price prediction,
- temperature forecasting,
- travel-time estimation,
- risk score prediction.

Unsupervised Learning

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Discover structure without explicit labels

In **unsupervised learning**, the dataset contains **inputs only**:

$$D = \{x_i\}_{i=1}^m$$

The model searches for **useful structure**, such as clusters, latent variables, or compressed representations.

Method family	Typical question	Example output
Clustering	Which examples are similar?	Groups of customers or documents
Dimensionality reduction	Can data be represented with fewer variables?	A low-dimensional visualization
Representation learning	Can the model learn features automatically?	Embeddings or latent vectors

A clustering algorithm may discover **groups with similar behavior** without being told the names of those groups in advance.

Reinforcement Learning

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Learning from interaction and reward

Reinforcement Learning is used when decisions occur over time and each action influences **future states**.

A **policy** maps states to actions:

$$\pi : S \rightarrow A$$

The agent receives **rewards** and tries to maximize **expected cumulative reward**:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right]$$

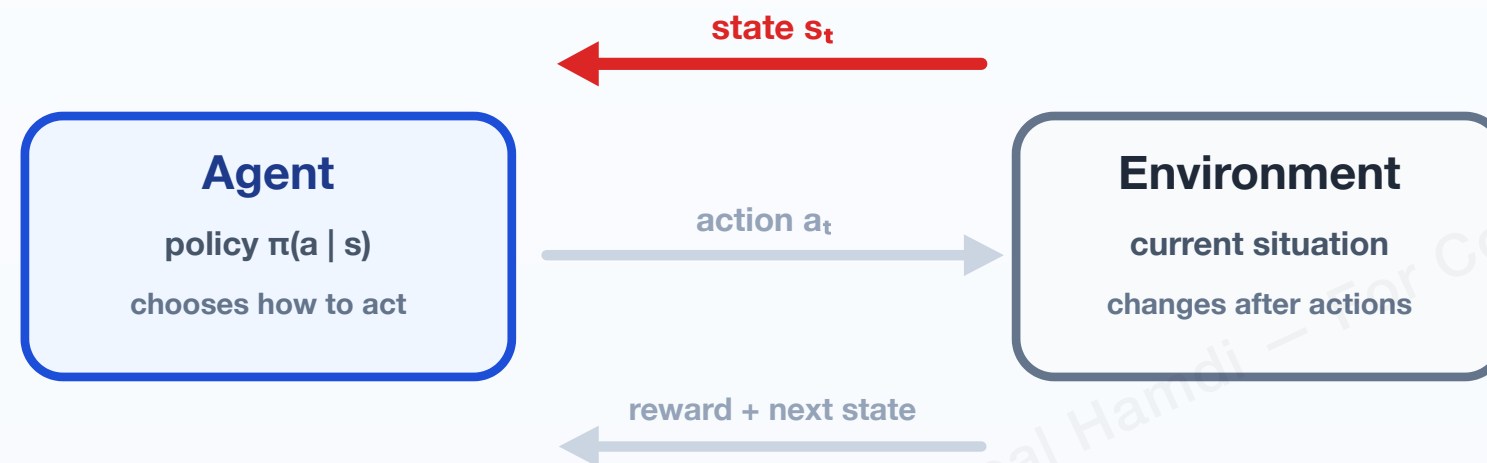
where γ is a **discount factor** that controls the importance of future rewards.

💡 Search plans by reasoning over a model. Reinforcement learning can improve behavior through **repeated interaction**, even when the correct action is **not explicitly labeled**.

Reinforcement Learning Interaction Loop

Learning through actions, rewards, and repeated experience

Reinforcement learning is a feedback loop between agent and environment



The agent first observes information about the current state of the environment.

Step 1 — observe the state

Teaching focus.

The environment provides the agent with a state or observation describing the current situation.

Input to agent

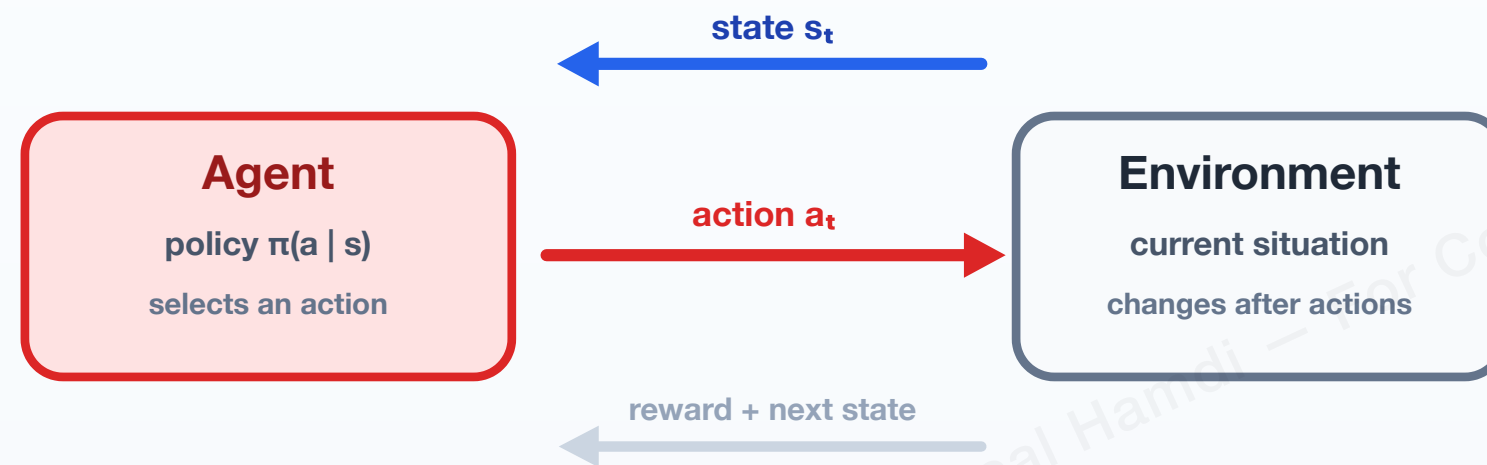
state s_t

Interpretation

what situation am I in?

Learning through actions, rewards, and repeated experience

The policy uses the state to choose an action



The action affects what happens next in the environment.

Step 2 — choose an action

Teaching focus.

The agent uses its current policy to select an action. Unlike supervised learning, it is not given the correct action as a label.

Decision rule

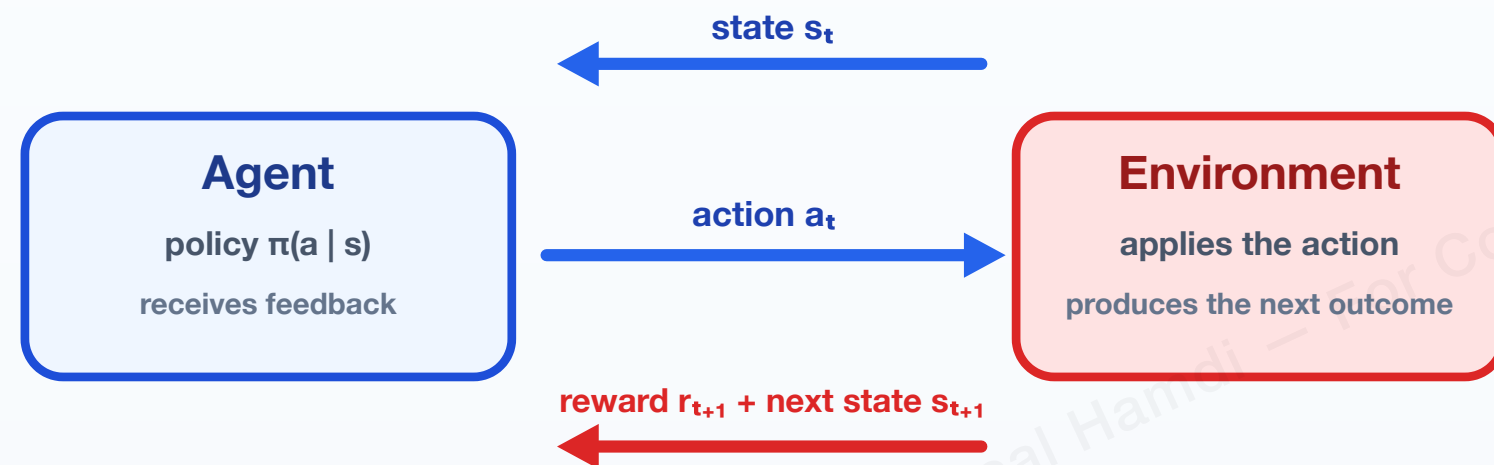
policy $\pi(a | s)$

Output

action a_t

Learning through actions, rewards, and repeated experience

The environment returns an outcome, not a labeled correct answer



Reward evaluates the outcome; the next state determines what the agent observes next.

Step 3 — receive reward and next state

Teaching focus.

After the action, the environment changes. The agent receives a reward signal and observes the next state.

Feedback

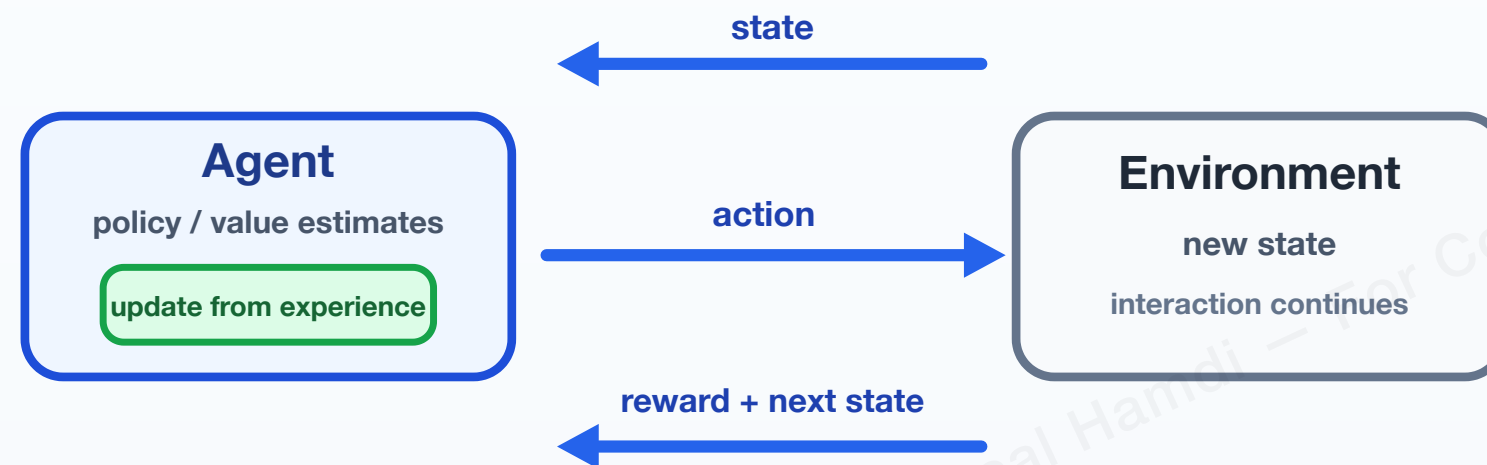
reward r_{t+1}

Transition

$s_t \rightarrow s_{t+1}$

Learning through actions, rewards, and repeated experience

Repeated experience can improve future decisions



observe → act → receive feedback → update → repeat

Goal: improve long-term cumulative reward, not merely the next immediate reward.

Step 4 — improve behavior and repeat

Teaching focus.

The experience can be used to update value estimates, the policy, or both. Repeated interaction should improve decisions that lead to greater long-term reward.

Learning signal

experience reward

Objective

maximize cumulative reward

Train, Validation, and Test Sets

Learning must be evaluated on unseen data

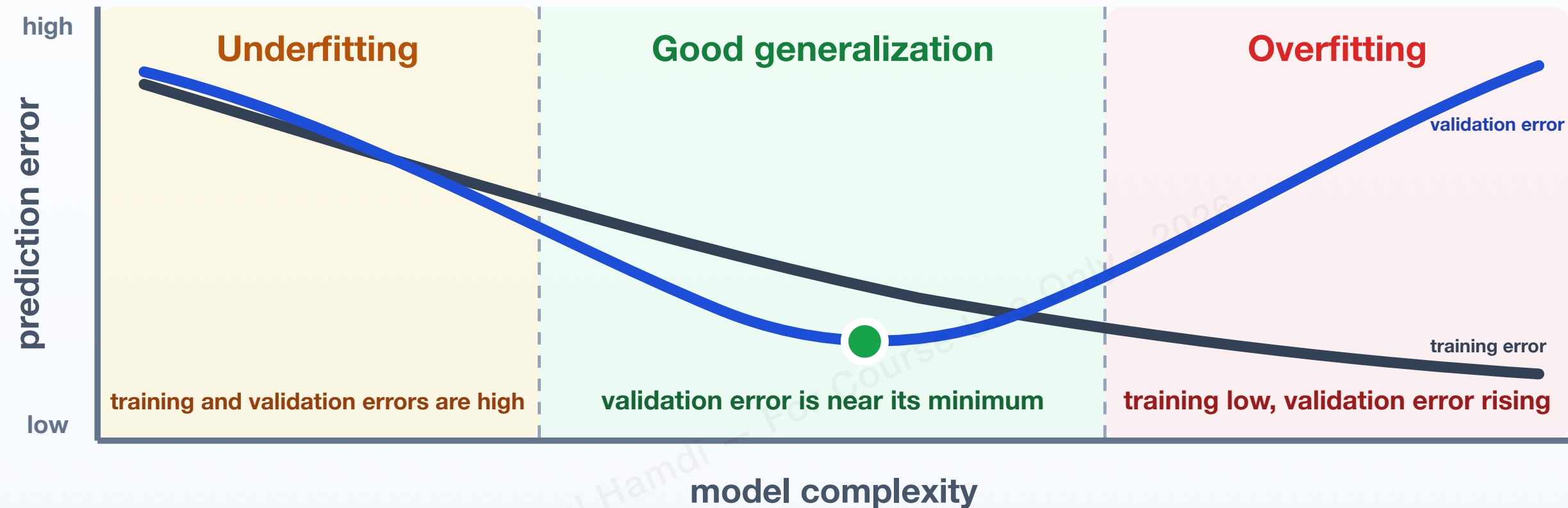
Split	Used for	Why it matters
Training set	Fit model parameters	The model directly learns from these examples.
Validation set	Choose model type, hyperparameters, and stopping point	Helps detect overfitting during development.
Test set	Estimate final performance	Should be used only after model selection.

A model that performs extremely well on training data may still fail on new examples. The real goal is **generalization**, not memorization.

Generalization and Overfitting

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Why training error alone is not enough



💡 **Key idea.** Increasing model complexity usually lowers training error, but validation error may begin to rise when the model starts fitting noise. Model selection should therefore use **validation performance**, while test data is reserved for final evaluation.

Decision Trees

A symbolic learning model

A **decision tree** predicts by testing features in a sequence of branches until it reaches a **leaf decision**.

```
def decision_tree_predict(example, tree):  
    if tree.is_leaf():  
        return tree.label  
  
    feature = tree.test_feature  
    value = example[feature]  
    branch = tree.child_for(value)  
    return decision_tree_predict(example, branch)
```

Decision trees are **interpretable** because a prediction can be explained as a **path of feature tests** from the root to a leaf.

A simple weather-based tree may predict whether to play tennis using features such as **outlook, humidity, and wind**.

Neural Networks

A flexible family of function approximators

A **neural network** applies layers of **weighted transformations** and nonlinear activation functions.

For one layer:

$$z = Wx + b$$

$$a = \sigma(z)$$

Stacking many layers allows the network to represent **complex functions**.

Why neural networks matter

- They can **learn representations automatically**.
- They scale well with **large datasets and computation**.
- Deep networks are central to modern **NLP and computer vision**.
- Their **interpretability and reliability** remain major challenges.

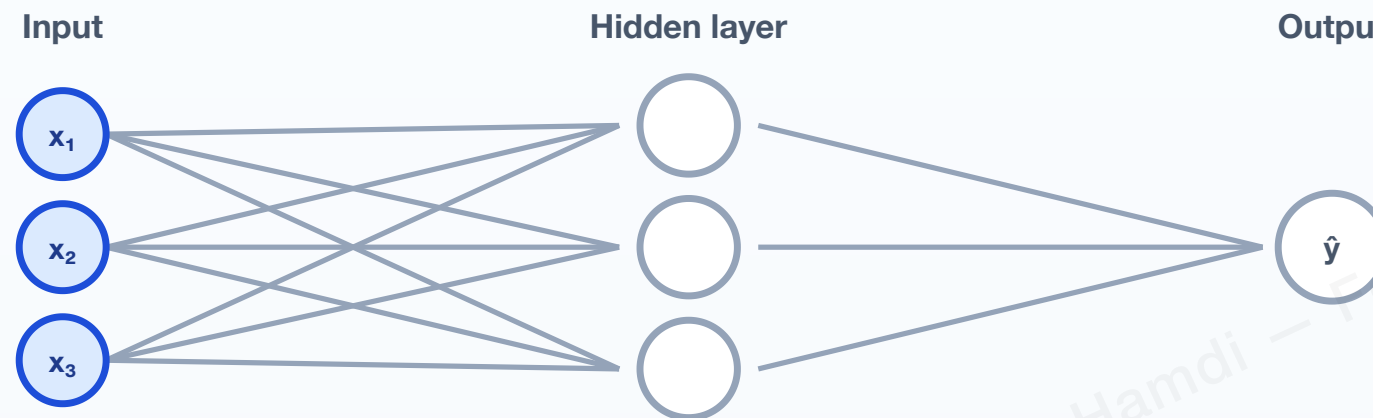
💡 **Deep learning** is not a separate goal from AI. It is a **modeling approach** used across many AI sub-domains.

How Neural Networks Learn

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Forward pass, loss, backpropagation, and weight updates

A neural network transforms inputs through connected layers



Connections have weights that control how strongly signals influence later units.

Step 1 — provide input features

Teaching focus.

Input values enter the network. Weighted connections determine how strongly each input contributes to units in the next layer.

Input

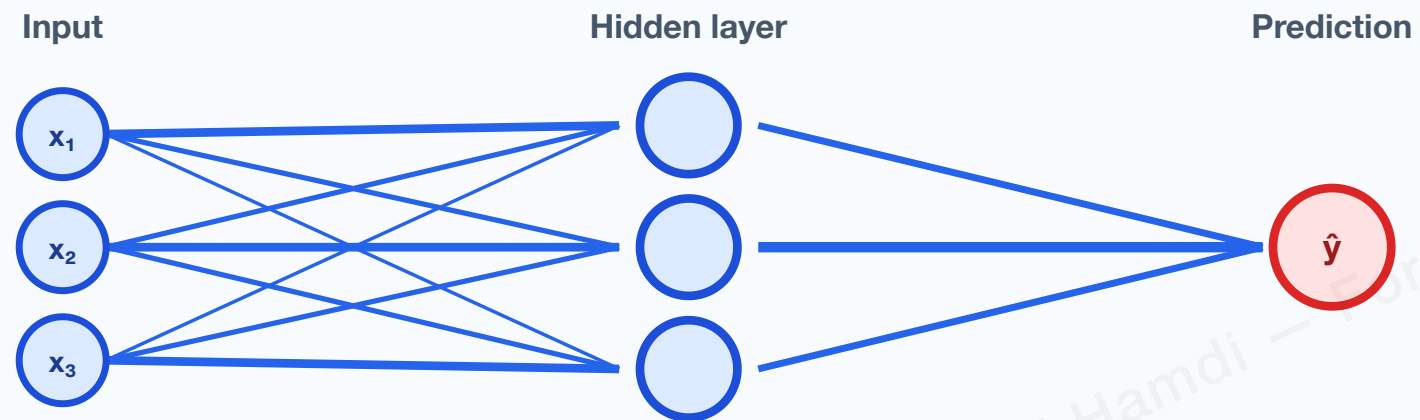
x_1, x_2, x_3

Learned parameters

weights

Forward pass, loss, backpropagation, and weight updates

A forward pass computes a prediction



input $\rightarrow$ weighted combinations $\rightarrow$ activations $\rightarrow$ prediction

Step 2 — perform a forward pass

Teaching focus.

Each unit combines weighted inputs and applies a nonlinear transformation. Information flows forward until the network produces a prediction.

Computation

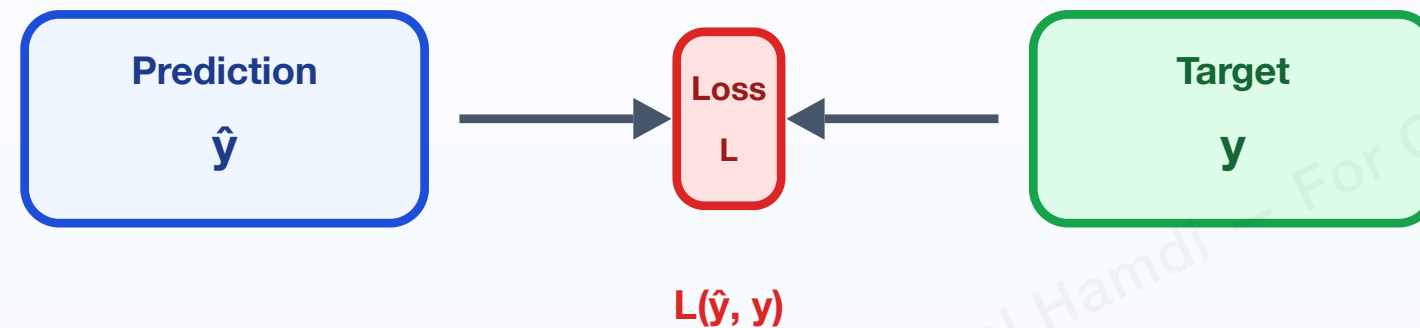
forward pass

Output

prediction $\hat{y}$

Forward pass, loss, backpropagation, and weight updates

Learning requires comparing prediction with the target



The loss measures how far the prediction is from the desired output.

Step 3 — compute the loss

Teaching focus.

During supervised training, the prediction is compared with the known target. A loss function converts the error into a numerical learning signal.

Compare

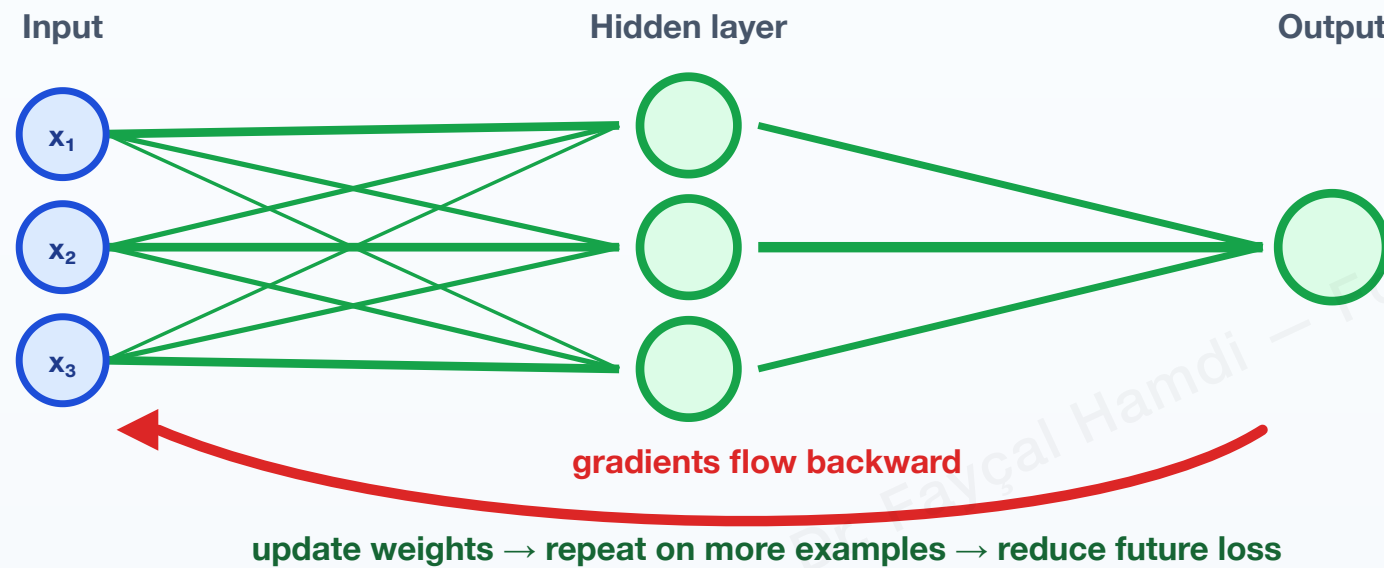
$\hat{y}$ versus y

Learning signal

loss $L(\hat{y}, y)$

Forward pass, loss, backpropagation, and weight updates

Backpropagation computes gradients; the optimizer updates the weights



Step 4 — update the weights

Teaching focus.

Backpropagation computes how the loss changes with each parameter. An optimizer then adjusts the weights, and training repeats over many examples.

Backward computation

gradients

Parameter change

optimizer update

Machine Learning Summary

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The key conceptual vocabulary

Concept	Meaning
Example	One data instance used for learning or evaluation
Feature / representation	Numerical description of an input
Model / hypothesis	Function used to make predictions
Loss	Error measure for one or more predictions
Training	Adjusting the model to reduce loss
Generalization	Performance on new, unseen examples
Overfitting	Learning noise or accidental patterns in the training data

Part II

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Natural Language Processing

Natural Language Processing studies how AI systems **represent, interpret, and generate** human language.

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Why NLP Is Difficult

Language is structured, ambiguous, and context-dependent

Language is challenging because the same words can have **different meanings depending on context**, world knowledge, speaker intention, and previous discourse.

Challenge	Example
Ambiguity	“I saw the man with a telescope.”
Context dependence	“It is too hot here.” may be a complaint or a request.
Compositionality	Sentence meaning depends on how words combine.
Long-range dependencies	A word may depend on another word many positions earlier.
World knowledge	Understanding often requires facts not explicitly stated.

💡 NLP is **not only word matching**. It requires representations that connect **form, meaning, context, and task objectives**.

NLP Processing Pipeline

From raw text to task output

A common pipeline is:

Text → Tokens → Representations → Model → Output

Examples of outputs include:

- a class label, such as sentiment or topic;
- a generated sentence;
- an extracted answer span;
- a translation;
- a summary;
- a ranked list of relevant documents.

The main technical step is turning language into **representations that preserve useful information** for the task.

Tokens and Vocabularies

Text must be segmented before modeling

Tokenization divides text into units such as **words, subwords, or characters**.

For a sentence

"AI systems learn from data"

a word-level tokenizer may produce

[AI, systems, learn, from, data]

A **vocabulary** maps each token to an **identifier**:

token $\rightarrow$ integer id

Tokenization choices affect **downstream behavior**. Unknown words, morphology, spelling variants, and multilingual text all make tokenization nontrivial.

Bag-of-Words Representation

A simple but limited representation

A **bag-of-words** model represents a document by **counting token occurrences** and **ignoring order**.

If the vocabulary is

$$V = [\text{AI}, \text{data}, \text{learn}, \text{search}]$$

then a document may be represented as a vector such as

$$x = [2, 3, 1, 0]$$

Bag-of-words can work surprisingly well for topic classification or spam detection, but it **loses word order and sentence structure**.

Word Embeddings

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Dense vectors that encode similarity

An **embedding** maps each token to a **dense vector**:

$$e : V \rightarrow \mathbb{R}^d$$

The purpose is to place **semantically or syntactically related words** near each other in vector space.

An embedding is a **learned numerical representation** that can carry useful information about word usage and context.

Intuition

Words such as “teacher,” “student,” and “lecture” may appear in related contexts, so their vectors may become **close** under a suitable training objective.

Language Modeling

Predict the next token from context

A **language model** assigns probabilities to token sequences. A common objective is **next-token prediction**:

$$P(w_t \mid w_1, w_2, \dots, w_{t-1})$$

For a sequence of length T , the probability can be decomposed as

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t \mid w_1, \dots, w_{t-1})$$

💡 Modern generative language models use this principle at very large scale, but the basic idea remains **probabilistic prediction over sequences**.

Attention Mechanism Intuition

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Context-sensitive language representation

Attention lets a token use information from other tokens

The

robot

opened

the

door

Each token is converted into a numerical representation before attention is applied.

Step 1 — represent the tokens

Teaching focus.

The sentence is divided into tokens, and each token is mapped to a numerical representation that the model can process.

Input

tokens

NLP challenge

meaning depends on context

Context-sensitive language representation

Choose one token whose contextual representation will be updated

The

robot

opened

query token

the

door

Step 2 — use “opened” as the query

Teaching focus.

For the token “opened,” the model compares its query with information associated with the other tokens to determine what context is relevant.

Query token

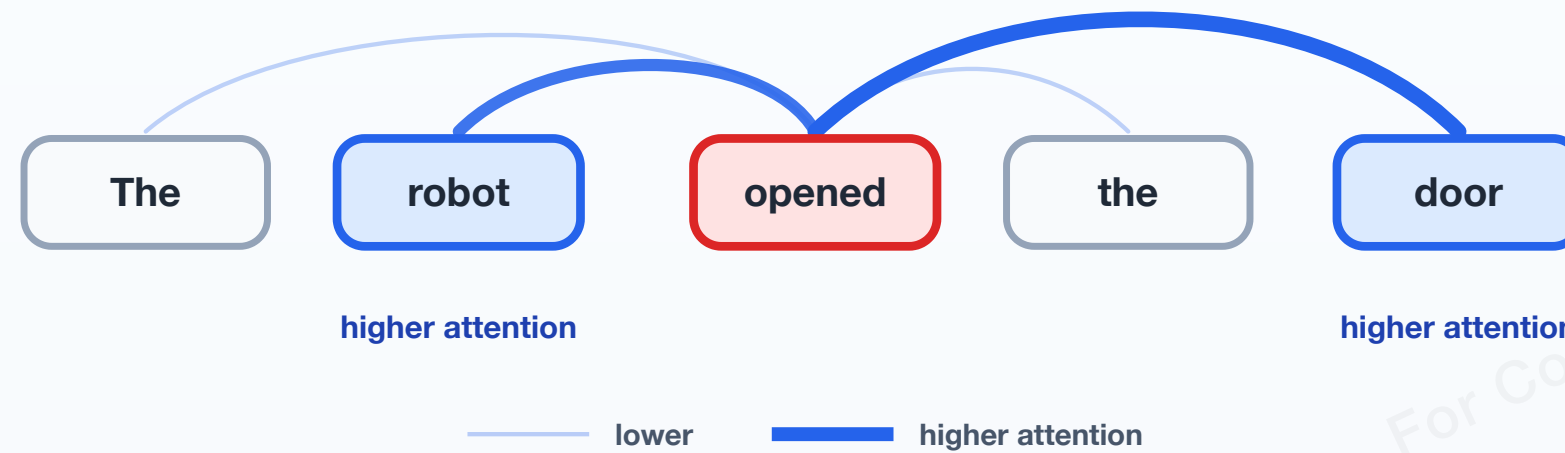
opened

Question

which tokens are relevant?

Context-sensitive language representation

Attention assigns different relevance weights to the context



Step 3 — compute attention over context

Teaching focus.

The query can assign different weights to all tokens. Here, “robot” and “door” receive stronger attention, while other words contribute less.

Important idea

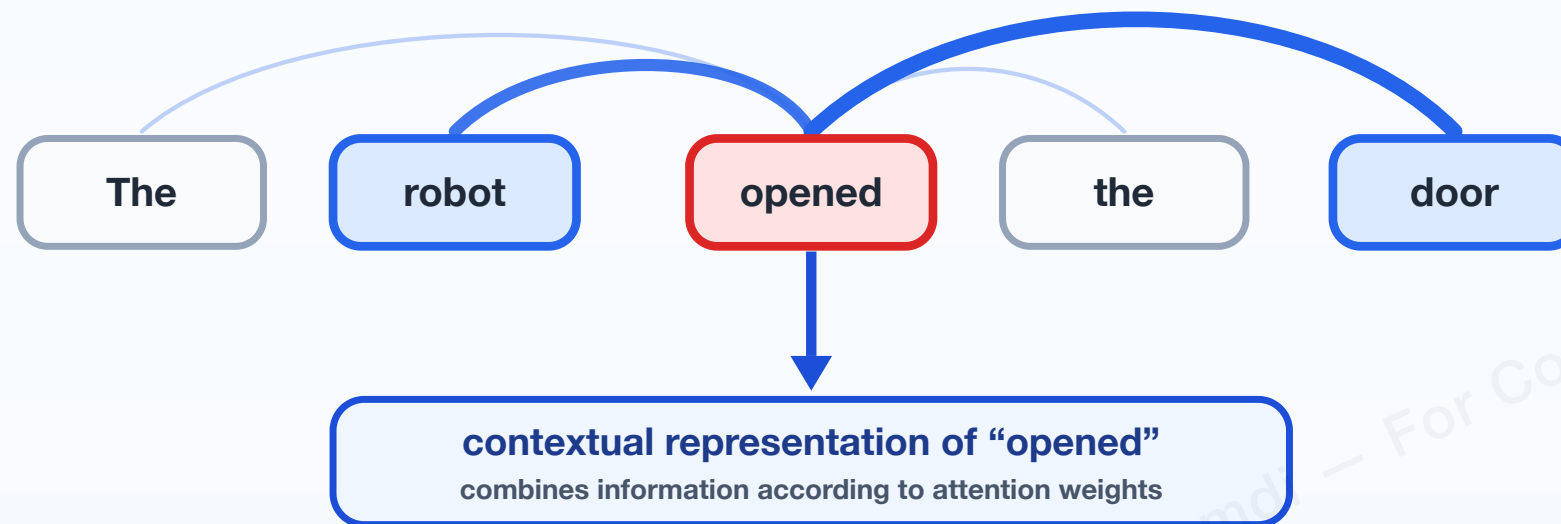
weighted relevance

Visual rule

thicker link = more attention

Context-sensitive language representation

Weighted context produces a context-sensitive representation



Step 4 – form a contextual representation

Teaching focus.

The model combines information from the attended token representations to produce a context-sensitive representation of “opened.”

Output

contextual vector

Can support

translation

question answering

classification

NLP Tasks

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Different tasks use different outputs

Task	Input	Output	Example use
Text classification	Document or sentence	Class label	Spam detection, sentiment analysis
Information extraction	Text	Entities or relations	Extract names, dates, organizations
Question answering	Question and context	Answer	Reading comprehension systems
Machine translation	Text in one language	Text in another language	Translation tools
Summarization	Long document	Shorter text	News or report summaries
Dialogue	Conversation history	Next response	Chatbots and assistants

NLP Limitations and Risks

Fluent language is not the same as grounded understanding

NLP systems can produce **impressive outputs**, but they also have important **limitations**.

Technical limitations

- sensitivity to **prompts and phrasing**,
- difficulty with long-term **factual consistency**,
- weak **grounding in physical reality**,
- vulnerability to ambiguous or adversarial input.

Social and ethical risks

- **biased or harmful outputs**,
- **privacy leakage**,
- **misinformation generation**,
- **overreliance** by users.

In applications that affect people, NLP output should be **evaluated, monitored, and interpreted carefully**.

Part III

Computer Vision

Computer Vision studies how AI systems extract **meaningful information** from images and video.

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What Is Computer Vision?

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From pixels to semantic interpretation

An image is a **numerical array**. A color image can be represented as

$$I \in \mathbb{R}^{H \times W \times C}$$

where H is height, W is width, and C is the number of channels.

The goal of computer vision is to infer **high-level information** such as:

- what **objects** are present,
- where objects are **located**,
- how objects relate **spatially**,
- what **action or scene** is occurring,
- how a robot should act based on visual input.

💡 The central challenge is that **pixels are low-level measurements**, while useful decisions require **high-level meaning**.

Images as Structured Data

Spatial relationships matter

Unlike a simple table row, image data has important **spatial structure**:

- nearby pixels often belong to the same **edge, region, or object**;
- patterns can appear at different positions;
- objects may vary in scale, orientation, illumination, and background;
- **partial occlusion** can hide important parts of an object.

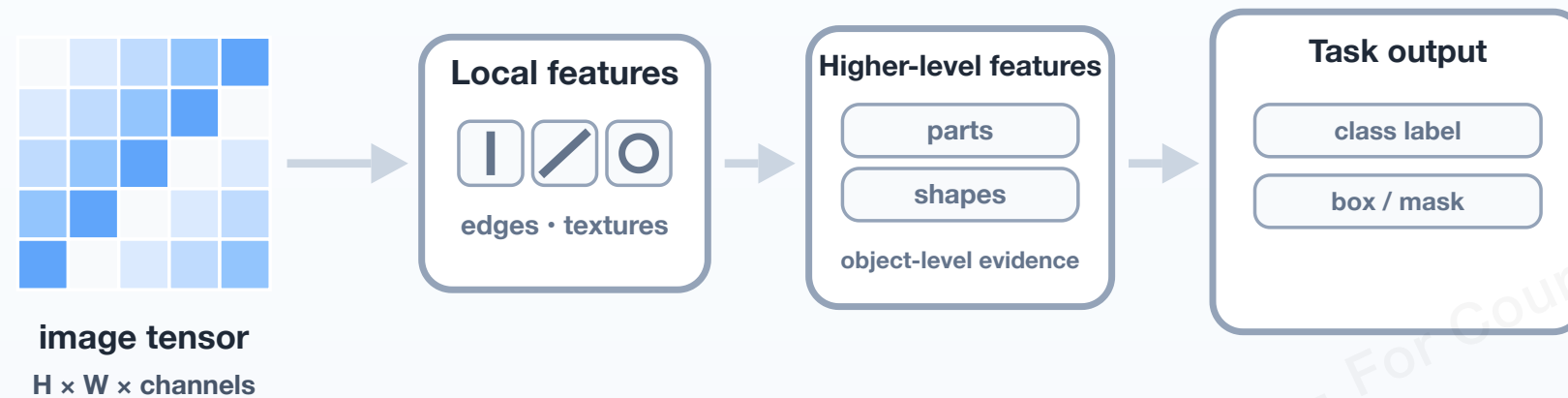
A good vision model should exploit **local structure** while building increasingly **abstract representations**.

A few pixels may form an **edge**, edges may form a **contour**, contours may form a **shape**, and shapes may support recognition of an **object**.

Computer Vision Pipeline

A CNN-style path from pixels to semantic information

One common vision pipeline builds meaning from numerical pixel values



Step 1 — pixels as numerical input

Teaching focus.

An image is represented as a structured tensor of pixel values. At this stage, the numbers describe intensity or color, not semantic objects.

Input

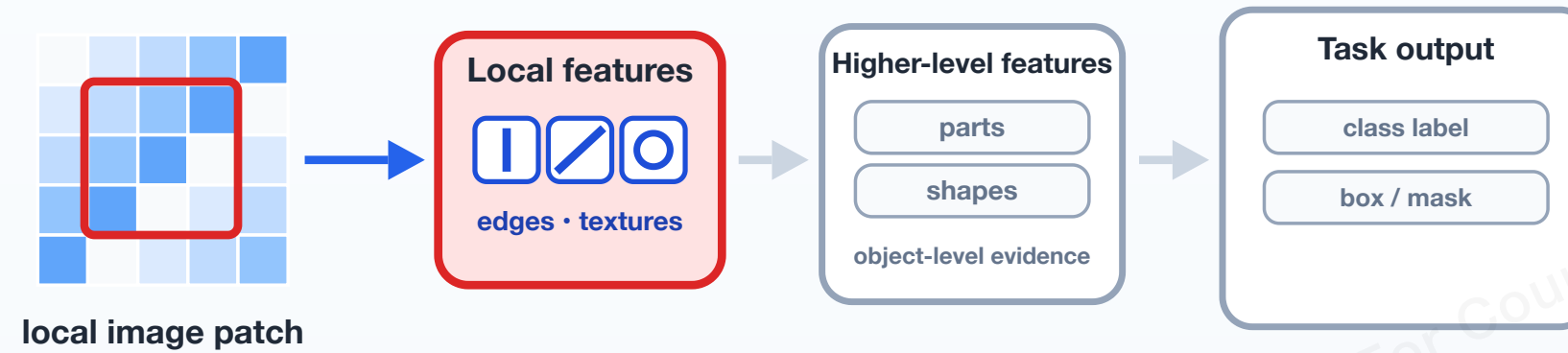
height × width × channels

Core challenge

extract meaning from pixels

A CNN-style path from pixels to semantic information

Learned filters respond to local visual patterns



Step 2 — learn local visual features

Teaching focus.

Learned convolutional filters respond to local patterns such as edges, colors, textures, and simple shapes. Their responses form feature maps.

Operation

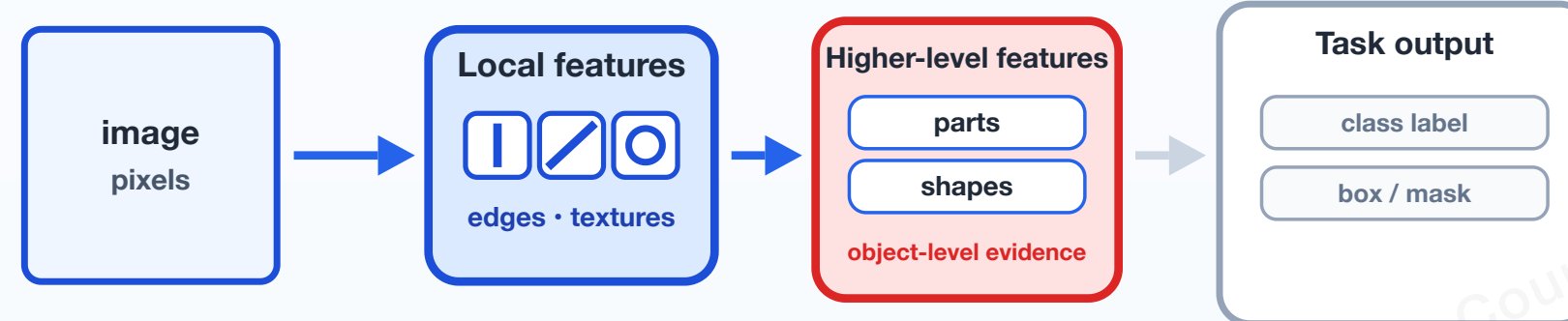
convolution

Early features

edges textures

A CNN-style path from pixels to semantic information

Deeper layers combine local responses into more abstract visual evidence



Step 3 — build a feature hierarchy

Teaching focus.

Deeper layers combine simpler local features into increasingly abstract representations such as parts, shapes, and evidence for whole objects.

Progression

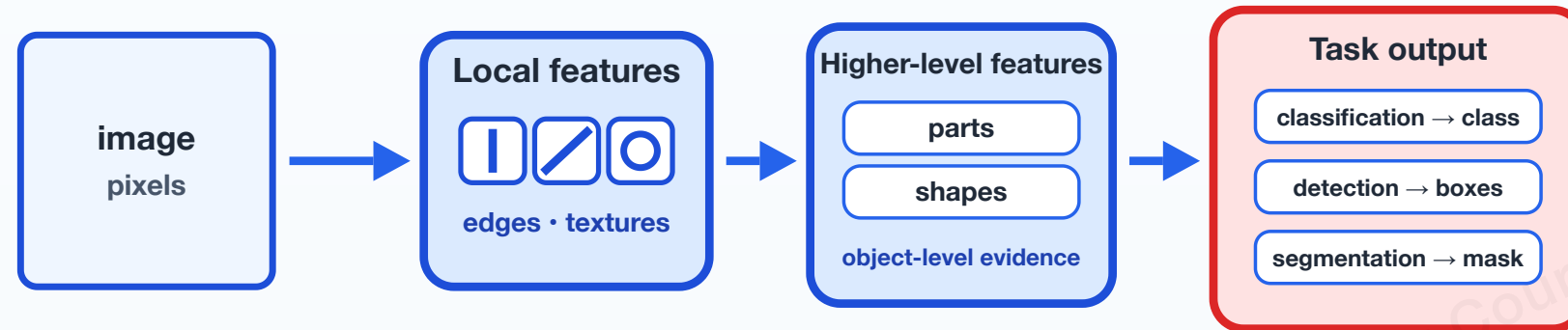
edges parts objects

Learning idea

hierarchical representation

A CNN-style path from pixels to semantic information

The learned representation is converted into a task-specific visual output



Step 4 — produce a task-specific output

Teaching focus.

The final representation is mapped to the output required by the task: a class label, object locations, segmentation masks, or another semantic interpretation.

Examples

classification detection segmentation

Result

semantic information

Convolution

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Local filters detect repeated patterns

A **convolutional filter** slides across an image and computes **local responses**.

For a simplified two-dimensional image I and filter K , convolution can be written as

$$(I * K)(i, j) = \sum_m \sum_n I(i + m, j + n)K(m, n)$$

A convolutional layer learns filters that respond to useful **local patterns**, such as edges, textures, or parts of objects.

Why convolution is useful

- It **shares parameters** across image locations.
- It exploits **local spatial structure**.
- It produces **feature maps** that preserve spatial organization.

Convolutional Neural Networks

Hierarchical feature learning

A **CNN** usually applies several layers of convolutions, nonlinearities, and pooling or downsampling operations.

A simplified structure is:

Image → low-level features → mid-level features → high-level features → prediction

Early layers may detect **edges**. Middle layers may detect **textures or object parts**. Later layers may support **object-level decisions**.

💡 Deep learning made computer vision more powerful by learning **feature hierarchies directly from data**.

Major Computer Vision Tasks

The output format depends on the task

Task	Question	Output
Image classification	What is in the image?	A class label or class probabilities
Object detection	What objects are present and where are they?	Class labels plus bounding boxes
Semantic segmentation	What class does each pixel belong to?	A label map over pixels
Instance segmentation	Which individual object does each pixel belong to?	Separate masks for object instances
Pose estimation	Where are key body or object points?	Keypoint coordinates
Visual question answering	What is the answer to a question about an image?	Text or class answer

Evaluation in Vision

Metrics must match the task

Task	Common metric idea	Interpretation
Classification	Accuracy, precision, recall, F1-score	How well labels are predicted
Detection	Intersection over Union, average precision	How well boxes match objects
Segmentation	Pixel accuracy, IoU, Dice score	How well predicted masks match true regions
Ranking or retrieval	Top- k accuracy, mean average precision	Whether correct items appear near the top

A high benchmark score does not guarantee **robust real-world performance**, especially under **distribution shift**.

Vision Challenges

Visual recognition is sensitive to context and conditions

Data challenges

- lighting variation,
- viewpoint changes,
- background clutter,
- occlusion,
- small objects,
- rare or underrepresented classes.

Deployment challenges

- real-time constraints,
- safety-critical decisions,
- sensor noise,
- privacy concerns,
- robustness to unexpected environments.

💡 Computer vision is not only about **recognizing objects**; it is also about building **reliable perception systems** for action, diagnosis, monitoring, and interaction.

Part IV

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Synthesis: Categorizing Advanced AI Domains

Machine Learning, NLP, and Computer Vision share many **technical ideas**, but they differ in **data type, representation, output, and evaluation**.

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Comparing the Three Domains

Same learning principles, different data and outputs

Domain	Typical input	Typical representation	Common output	Example applications
Machine Learning	Tables, signals, text, images, interactions	Features, embeddings, learned parameters	Class, number, cluster, policy	Prediction, recommendation, anomaly detection
NLP	Text or speech transcripts	Tokens, embeddings, contextual vectors	Label, answer, translation, generated text	Chatbots, search, summarization, sentiment analysis
Computer Vision	Images or video	Pixels, feature maps, visual embeddings	Class, bounding box, mask, caption	Medical imaging, autonomous driving, surveillance, robotics

Machine Learning is both a **sub-domain** and an **enabling methodology** for other sub-domains such as NLP and Computer Vision.

Common Deep Learning Pattern

Representation learning across domains

Many modern AI systems follow a similar structure:

raw input $\rightarrow$ learned representation $\rightarrow$ task-specific output

Examples:

- text $\rightarrow$ **token embeddings** $\rightarrow$ answer or summary,
- image $\rightarrow$ **feature maps** $\rightarrow$ object detection,
- user behavior $\rightarrow$ **latent vector** $\rightarrow$ recommendation,
- state observation $\rightarrow$ **policy representation** $\rightarrow$ action.

💡 The domain changes the input and task, but the core problem remains: **learn useful representations** that support good decisions.

Multimodal AI

Combining language, vision, and other data sources

A **multimodal AI system** processes more than one type of data.

Examples include:

- image captioning: image input, text output;
- visual question answering: image plus question, answer output;
- speech assistants: audio input, text and action output;
- robotics: vision, language instructions, sensors, and actions.

A **multimodal representation** tries to connect information from different modalities into a shared or coordinated model.

A robot that follows the command “pick up the red cup” must connect **language, visual perception, object localization, and action planning**.

Responsible Use Across AI Domains

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Capabilities must be matched with limitations

Concern	Why it matters
Bias	Models may reproduce unfair patterns present in data.
Privacy	Training and deployment data may contain sensitive information.
Robustness	Models can fail under distribution shift or adversarial conditions.
Explainability	Stakeholders may need reasons for predictions or decisions.
Safety	Errors can cause harm in medical, transport, legal, or security contexts.
Accountability	Human responsibility remains necessary even when systems are automated.

Advanced AI should be evaluated not only by **accuracy**, but also by **reliability, fairness, security, transparency, and suitability** for the deployment context.

Worked Categorization Activity

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Identify the domain, input, output, and learning setting

For each system, classify it according to the table below.

System	Likely domain	Input	Output	Learning setting
Detecting pneumonia from chest X-rays	?	?	?	?
Translating English text into French	?	?	?	?
Predicting house prices from property features	?	?	?	?
Recommending videos to users	?	?	?	?
A robot learning to navigate by reward feedback	?	?	?	?

Expected insight: Many systems combine domains. For example, medical image diagnosis is Computer Vision plus Machine Learning, and robotics may combine Vision, NLP, planning, and Reinforcement Learning.

Suggested classification

System	Likely domain	Input	Output	Learning setting
Detecting pneumonia from chest X-rays	Computer Vision + ML	Chest X-ray image	Disease class or probability	Supervised learning — classification
Translating English text into French	NLP + ML	English text / token sequence	French text / token sequence	Supervised learning — sequence-to-sequence
Predicting house prices from property features	Machine Learning	Numerical and categorical property features	Predicted house price	Supervised learning — regression
Recommending videos to users	ML / Recommender Systems	User behavior, preferences, and video features	Ranked videos or relevance scores	Supervised / implicit-feedback learning
A robot learning to navigate by reward feedback	Robotics + Reinforcement Learning	Sensor observations / current state	Navigation action	Reinforcement learning

Classification principle: identify **what information enters the system**, **what decision or prediction it produces**, and **what form of feedback is available during learning**.

💡 A real AI application may belong to **multiple domains simultaneously**. The domain describes *what kind of problem is being solved*, while the learning setting describes *how the system improves from experience or data*.

Mini-Quiz

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Check your understanding

1. What is the difference between supervised and unsupervised learning?
2. Why is test-set performance more important than training-set performance?
3. What does an embedding represent in NLP?
4. What is the basic objective of a language model?
5. Why are convolutions useful for image processing?
6. Give one example of a multimodal AI system.

Suggested answers: supervised learning uses target outputs; unsupervised learning discovers structure; test performance estimates generalization; embeddings are dense numerical representations; language models predict token probabilities; convolutions detect local visual patterns; image captioning is multimodal.

Lecture 6 Summary

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Key ideas to retain

Learning and language

- Machine Learning enables systems to improve from data.
- Supervised learning uses labeled examples.
- Unsupervised learning discovers structure.
- Reinforcement learning learns through rewards.
- NLP transforms language into numerical representations.
- Attention builds context-sensitive representations.

Vision and integration

- Computer Vision turns pixels into semantic information.
- CNNs learn hierarchical visual features.
- Vision tasks include classification, detection, and segmentation.
- Modern AI systems often combine learning, NLP, vision, and decision-making.
- Advanced AI must be evaluated for reliability and responsible use.

End-of-Lecture Check

Five questions to test your understanding

1. Why is Machine Learning important for tasks where explicit rules are difficult to write?
2. How do x_i , y_i , h_θ , and L relate in supervised learning?
3. Why can overfitting produce excellent training performance but poor real-world performance?
4. Why do NLP systems need tokenization and representations before modeling?
5. How is Computer Vision different from general tabular prediction?

Course-level synthesis: The course began with agents and search, then moved through constraint reasoning and adversarial reasoning. Lecture 6 connects these foundations to modern AI systems that learn from data, process language, and perceive visual environments.

Suggested Reading

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Primary textbook

S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th edition, Prentice Hall, 2020.

- Chapter 19: Learning from Examples
- Chapter 21: Deep Learning
- Chapter 23: Natural Language Processing
- Chapter 24: Deep Learning for Natural Language Processing
- Chapter 25: Computer Vision

💡 For this lecture, focus on the conceptual categories, formal learning vocabulary, and the relationship between data type, representation, model, and output.